# Queue

## Queue Implementation

Queue can be implemented using Array and Linked List.

1. **Queue implementation using Array**.

```
#include <stdio.h>

#include<conio.h>

  #define MAX 50

  void insert();

      void delete();

       void display();

       int queue_array[MAX];

        int rear = - 1;

       int front = - 1;

    void main()

     {

   int choice;

   while (1)

  {

   printf("1.Insert element to queue \n");

    printf("2.Delete element from queue \n");

    printf("3.Display all elements of queue \n");

            printf("4.Quit \n");

     printf("Enter your choice : ");
```

Ashwani

```
    scanf("%d", &choice);
     switch (choice)
     {
            case 1: insert();
      break;
              case 2: delete();
       break;
            case 3:  display();
        break;
        case 4: exit(1);
           default: printf("Wrong choice \n");
      }
   } }
void insert()
        {
   int add_item;
   if(rear == MAX - 1)
    {
  printf("Queue Overflow \n");
        }
   else if(front == - 1)
       {
        front=0;
        rear=0;
       }
```

Ashwani

```
    else
{
      printf("Inset the element in queue : ");
       scanf("%d", &add_item);
     rear = rear + 1;
      queue_array[rear] = add_item;
         front=0;
   }
}

void delete()
{
   if (front == - 1 || front > rear)
  {
      printf("Queue Underflow \n");
      return;
   }
  else
  {
       printf("Element deleted from queue is : %d\n", queue_array[front]);
      front = front + 1;
  }
}

 void display()
```

Ashwani

```
        {
  int i;
 if (front == - 1)
    printf("Queue is empty \n");
 else
            {
    printf("Queue is : \n");
   for (i = front; i <= rear; i++)
       printf("%d ", queue_array[i]);
   printf("\n");
  }
}
```

## 2. Queue implementation using linked list

```
#include<stdio.h>
#include<conio.h>

struct Node
{
  int data;
  struct Node *next;
}*front = NULL,*rear = NULL;
```

Ashwani

```
void insert(int);
void delete();
void display();

void main()
{
   int choice, value;
   clrscr();
   printf("\n:: Queue Implementation using Linked List ::\n");
   while(1)
{
      printf("1. Insert\n2. Delete\n3. Display\n4. Exit\n");
      printf("Enter your choice: ");
      scanf("%d",&choice);
      switch(choice){
            case 1: printf("Enter the value to be insert: ");
                  scanf("%d", &value);
                  insert(value);
                  break;
            case 2: delete(); break;
            case 3: display(); break;
            case 4: exit(0);
            default: printf("\nWrong selection!!! Please try again!!!\n");
```

Ashwani

```
    }
  }
}
void insert(int value)
{
  struct Node *newNode;
  newNode = (struct Node*)malloc(sizeof(struct Node));
  newNode->data = value;
  newNode -> next = NULL;
  if(front == NULL)
    front = rear = newNode;
  else{
    rear -> next = newNode;
    rear = newNode;
  }
  printf("\nInsertion is Success!!!\n");
}
void delete()
{
struct Node *temp;
  if(front == NULL)
    printf("\nQueue is Empty!!!\n");
  else{
```

Ashwani

```
    temp = front;
    front = front -> next;
    printf("\nDeleted element: %d\n", temp->data);
    free(temp);
  }
}
void display()
{
  if(front == NULL)
    printf("\nQueue is Empty!!!\n");
  else{
    struct Node *temp = front;
    while(temp->next != NULL){
        printf("%d--->",temp->data);
        temp = temp -> next;
    }
    printf("%d--->NULL\n",temp->data);
  }
}
```

Ashwani